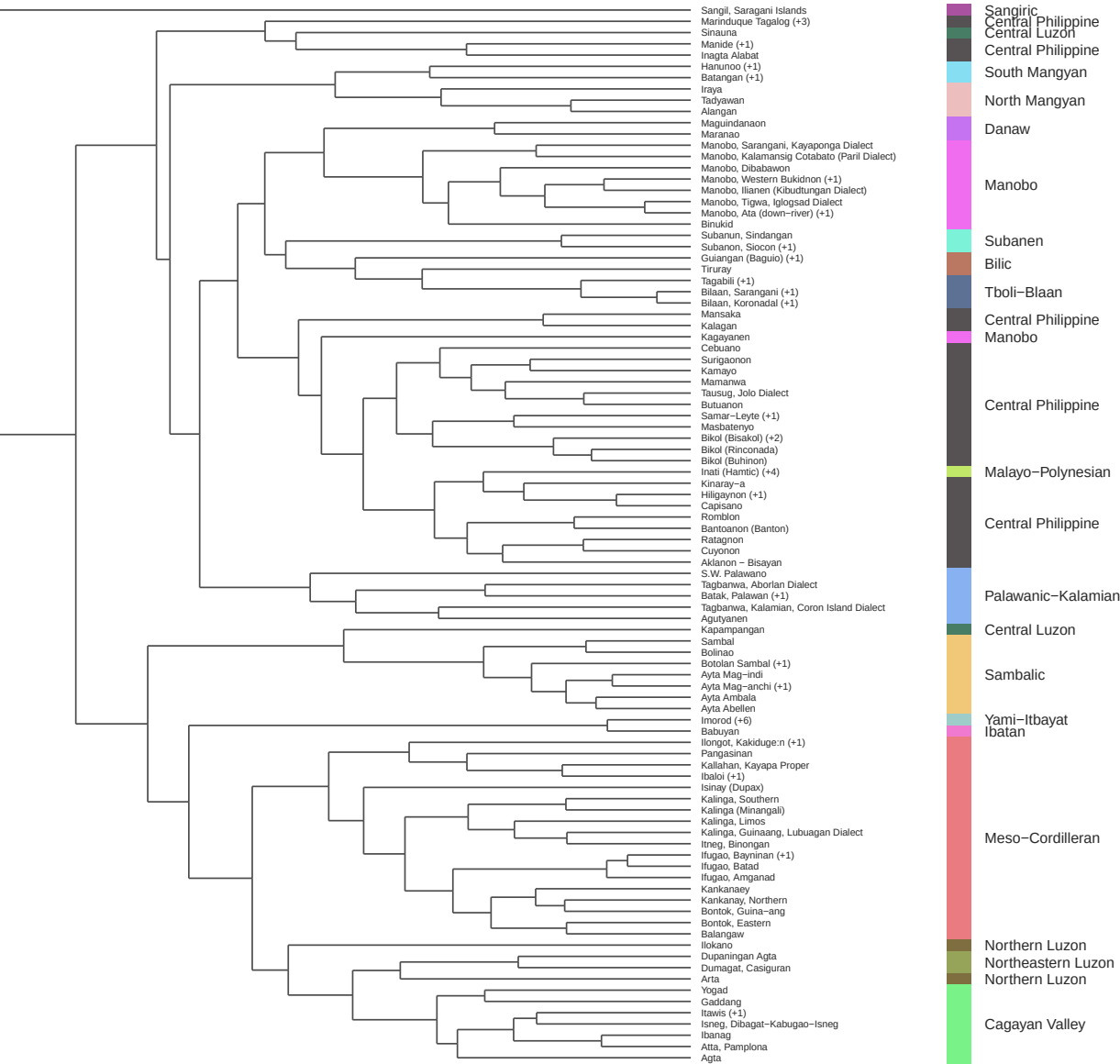
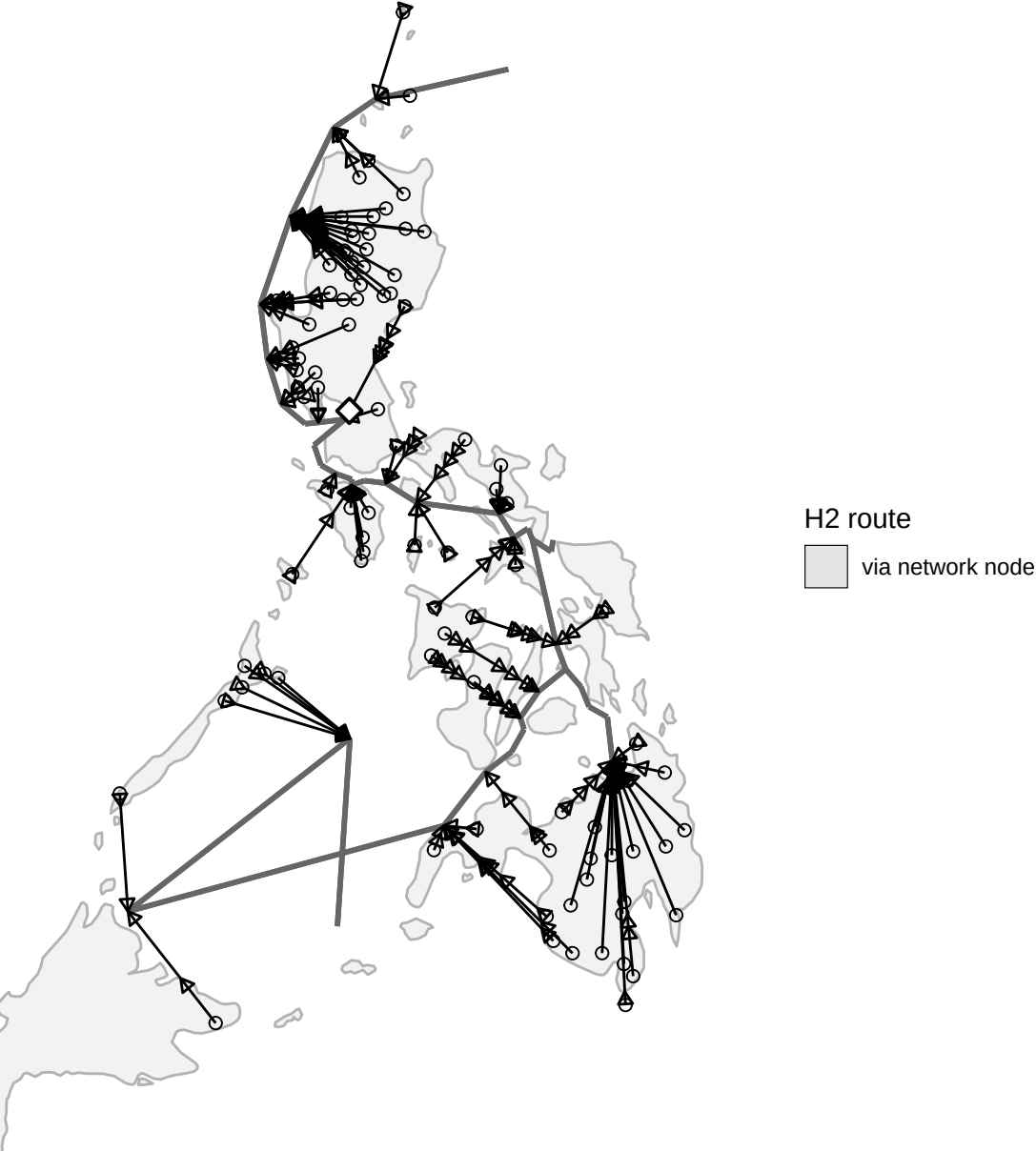
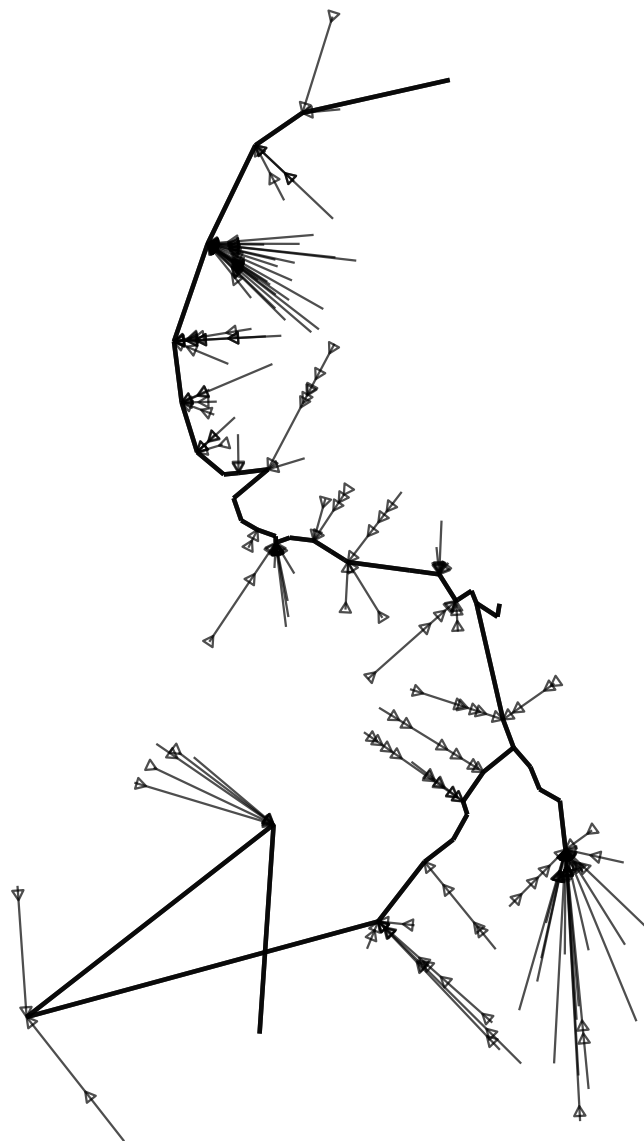


Philippine study languages (Cognate) – phylogeny (ABVD / King et al. 2024)



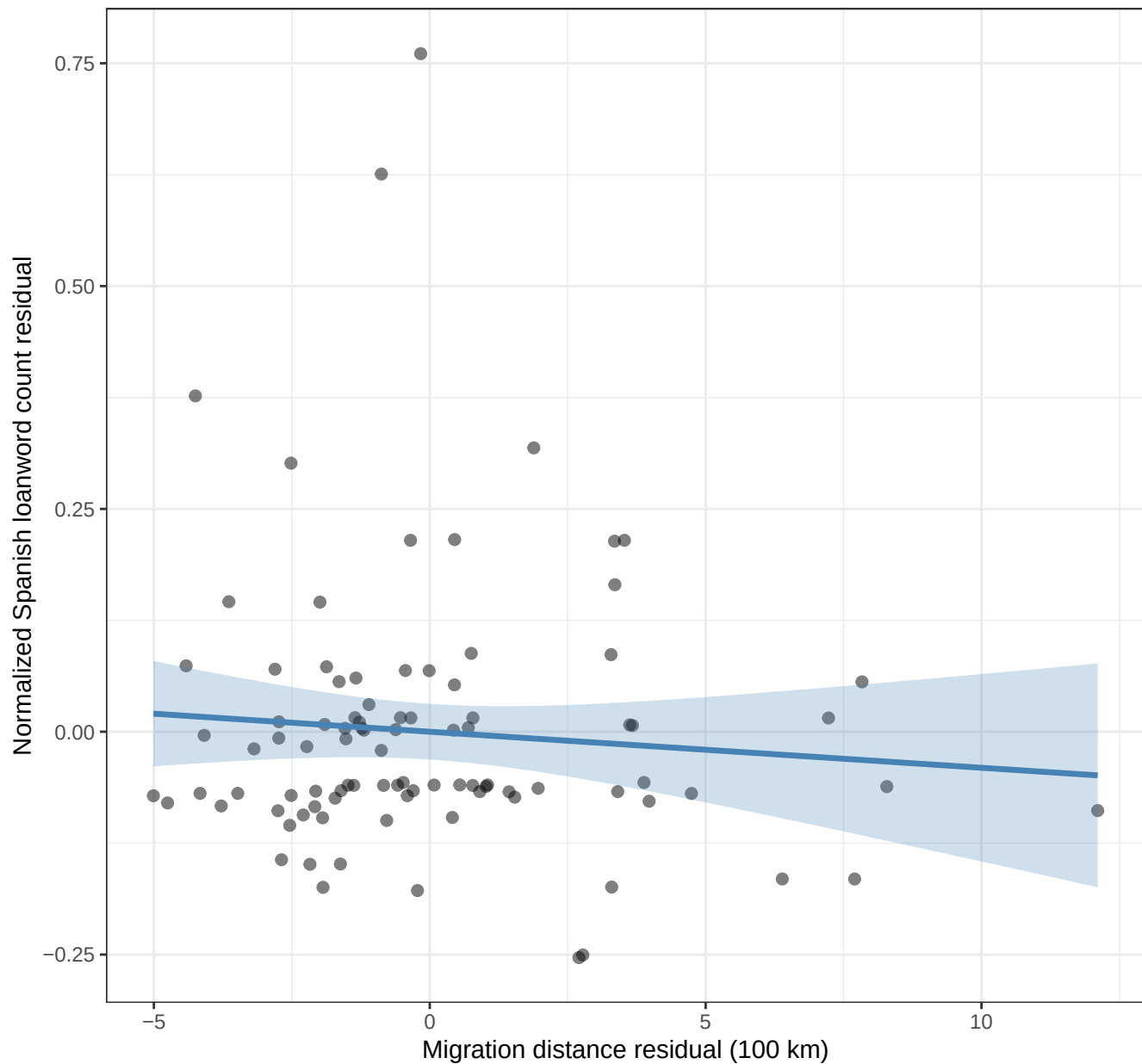
Cognate Historical Routes Waypoint System





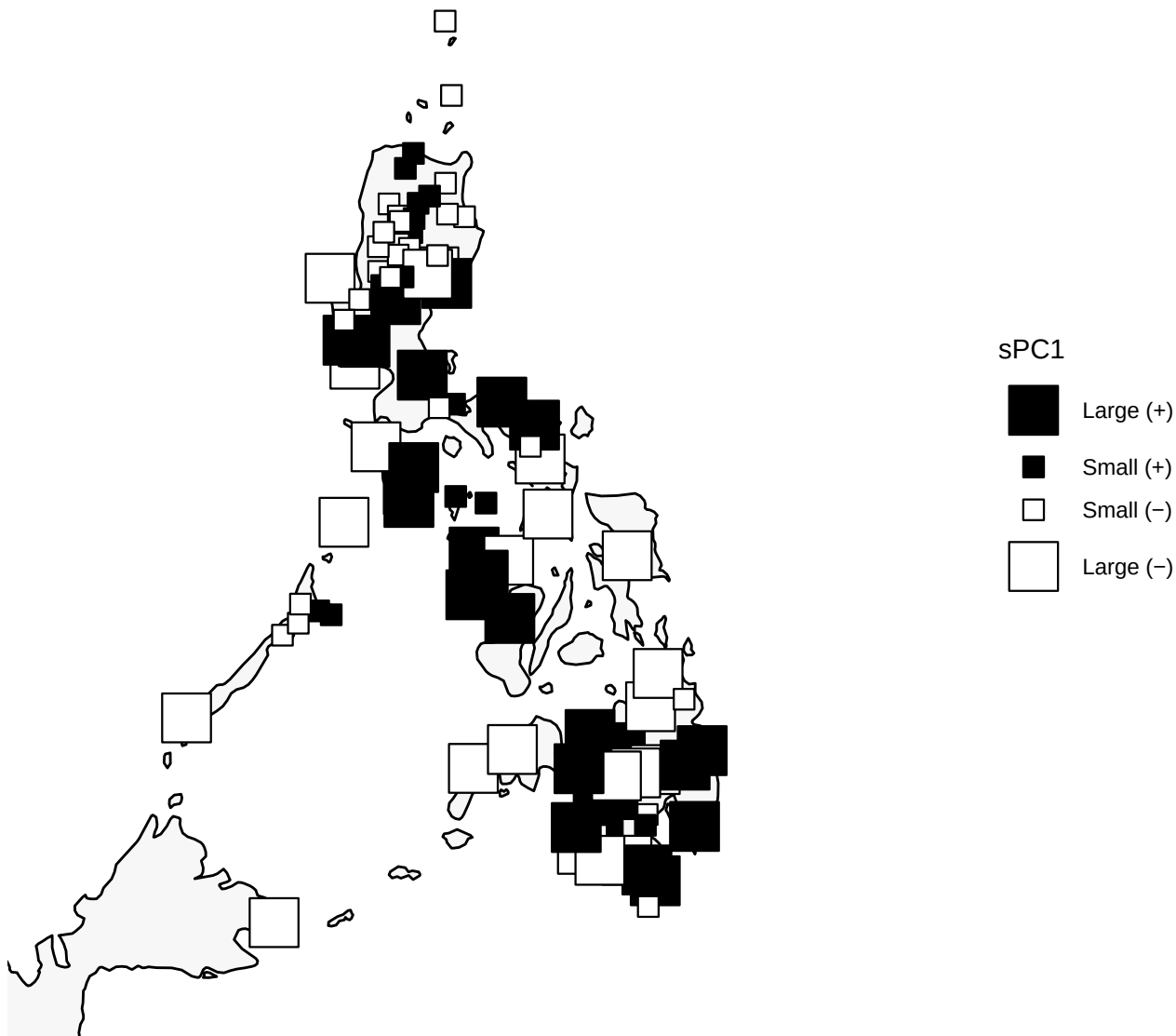
Spanish loanwords vs. migration distance, phylogeny removed

slope = -0.0040 per 100 km ($p = 0.431$), partial $r = -0.083$, 2 eigenvectors



Cognate PVR-sPCA: phylogeny-scrubbed spatial structure

Moran's I permutation $p = 0.438$ (1000 permutations) | eigenvector $R^2 = 0.078$



Cognate sPCA: phylogeny-scrubbed multivariate spatial structure

permutation p = 0.001 (1000 permutations) | variance explained = 0.032

